Mutriku wave power

Unpacking the full life-cycle footprint of breakwater wave energy

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Mutriku integrates oscillating water columns into a harbour breakwater in northern Spain. It combines electricity generation with turbine testing. This case asks whether improving output can justify the material, ecological and societal burdens of building, operating and eventually retiring the plant [1,4].

How oscillating water column technology evolved

Sources: [7,8]

1910 — Bochaux Praceique builds the believed first oscillating water column (OWC)

1940s — Masuda develops a wave-powered navigation buoy

Late 1970s — Wells turbine invented: airflow both ways, no rectifying valves needed

1985 — Toftestallen, Norway: shoreline OWC prototype

1999–2000 — Pico (Azores) and LIMPET (Scotland) plants; LIMPET ran over 10 years

2011 — **Grid connection: 16 chambers × 18.5 kW (296 kW) [1,14]**

Mutriku integrates the OWC into a breakwater, so one structure also protects the harbour.

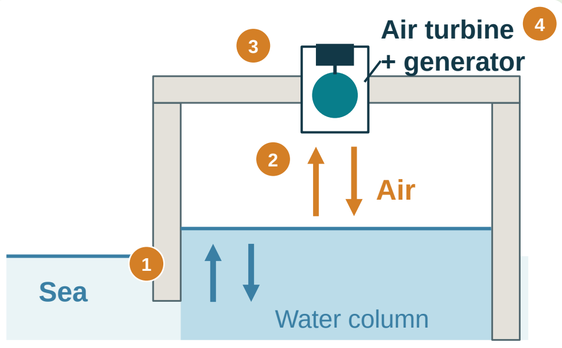
Did you know?

Europe's first: EVE describes Mutriku as the first commercial plant in Europe to generate electricity from waves [13].

Longest-running: EVE reports more operating hours since 2011 than any other wave plant of its kind [13].

Open to visitors: the plant is part of the tourist offer of the UNESCO-recognised Basque Coast Geopark [14].

01 How it works



Submerged opening

1. Wave lifts the water column
2. Air is pushed through the turbine
3. Wells turbine keeps turning one way
4. Generator makes electricity [1]

16 × 18.5 kW = 296 kW [1,14]

Simplified section, not to scale. Arrows show alternating motion. Diagram based on [1].

02 Where it sits: location and settings



Photo: Txo (2026), Wikimedia Commons, CC0 1.0

Location: inside the harbour breakwater at Mutriku, on the Basque coast of northern Spain.

Why there: built into the breakwater, one structure shares harbour protection, port use and electricity generation [1,11]. The BiMEP Mutriku area also hosts turbine testing [4,9].

03 Where the carbon burden sits

226 g CO_2e/kWh — Life-cycle total, 20-year model [1]

? Dismantling emissions not quantified (set to 0)



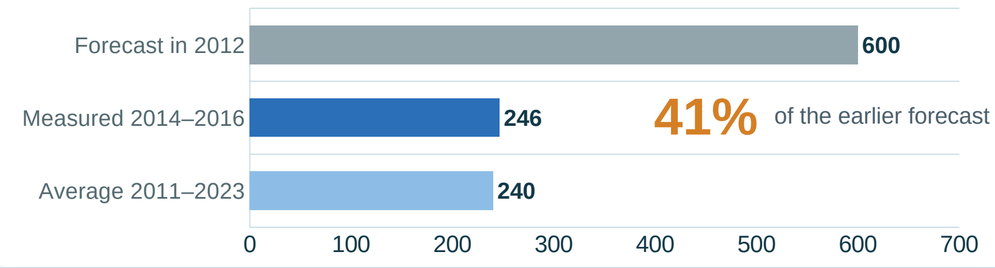
- **Construction 86.6%**
- **Operation & maintenance 5.5%**
- **End of life 7.9%**

Secondary inventories | Staff data | Dismantling set to 0 (no data)

Rounded stages sum to 225.3 vs 226 reported [1]. Model: 20 years, 4.93 GWh [1].

Construction dominates, so material demand is the priority. The model includes breakwater construction but excludes its maintenance [1]. Mutriku emits 53–94% more greenhouse gas than the other ocean devices compared in [1].

04 Output: forecast and evidence



41% of the earlier forecast

MWh/year [2,3]. Average: over 3 GWh [13,14] ÷ 12.5 years (author's calc.).

About 10 of the 14 turbines analysed working on average [2]

Cumulative generation reached 1 GWh in 2016 and 2 GWh in 2020 [17], then over 3 GWh by the end of 2023 [13,14], so the gap persists.

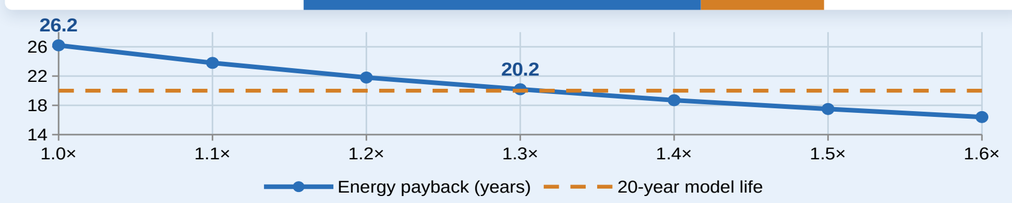
Beyond carbon: the wider footprint

▲ Favourable ▼ Adverse – Neutral

- ▼ Construction noise was heard 3–10 km away and worried locals [10]; one month of 2022 operating noise monitoring (one of three hydrophones failed) found no plant contribution to the soundscape [18]
- – Breakwater may alter sediment and flow; the OWC adds no significant change [10]
- ▲ Turbines sit in the air, not the sea, but salt exposure and splashing still cause corrosion [11,15]; plant sits on the breakwater, not the seabed [12]
- ▼ Most other LCA categories (e.g. marine ecotoxicity) are also driven by construction [1]
- ▼ Construction used about 12.9 kt of concrete and 21.3 kt of limestone [1]

05 Energy payback

Model life	20 years	
Payback	+6.2	26.2 years



Extend service life | **Raise net output** | **Use less material**

Break-even needs about 31% more output, or about 26 years' life (26.2 ÷ 20). [1] quotes 8 years for solar PV and 1.5 for wind. Axis: output relative to model; author's scaling, embodied energy fixed [1].

06 Impacts, trade-offs and mitigation

	Environment	Society
Build	▼ 195 g CO_2e/kWh [1]; hard-bottom habitat site, effects not assessed [1,10] ●	▼ Construction noise concerned local residents [10] ●
Operate	▲ Low emissions: 12.4 g CO_2e/kWh [1] ● ▲ No significant acoustic impact in 2016 monitoring; no complaints in ~10 years [10] ● ▼ Near-field sound and species effects unresolved; no rise seen ~1 km away [5,10] ●	▲ Electricity [1,14], turbine-testing access [4,9], harbour protection [11] ● ▲ Visitor interest suggests a local benefit [3,14] ● ▼ Public cost: €6.7M project, €2.3M for the energy plant [16] ●
Retire	? Habitat effects of removal not assessed; dismantling emissions modelled as zero [1] ● ▲ Steel recycling cuts metal depletion; more recycling is not always better overall [1] ●	? Not assessed here. Jobs, current acceptance unverified; 2013 study dated [6] ●
Fix	Cut virgin materials | Verify marine durability | Upgrade turbines [4] | Compare end-of-life options	Coordinate work schedules | Track output and complaints | Residents shape decisions [5,6]

● Modelled ● Source-reported ● Unsourced or unresolved

Community milestones

Selected milestones, not a time-scaled axis.

- **Jul 2011 – Plant opens [3]**
- **Dec 2011 – Storm shutdown [3]**
- **Mar 2012 – Operation resumes [3]**
- **2013 – Social study [6]**
- **2025 – SafeWAVE paper [5]**

Selection prioritised recent (2018+) primary studies on impacts, output and turbines; pre-2018 sources [3,6,11,16] are used only where no newer Mutriku-specific study was found. This focused review is not systematic.

Conclusions and Recommendations

Key findings

1. **Construction dominates:** 86.6% of the 226 g CO_2e/kWh life-cycle total [1].
2. **Payback exceeds service life:** 26.2 years against a 20-year model life; output was ~41% of forecast [1–3].
3. **Evidence is incomplete:** dismantling is modelled as zero [1]; social evidence is dated [6].

Key judgement

Wave electricity alone does not make Mutriku sustainable. It is promising, but only lower material use, higher output or longer life, with full life-cycle accounting and fair sharing of burdens, can justify it.

1. Upgrade within the existing structure
2. Compare equivalent alternatives
3. Test lifetime, downtime, recycling
4. Approve only with full accounting
5. Publish monitoring; revisit

References (IEEE style)

[1] R. León Lira, L. Güereca and J. Lekube, "Life cycle assessment of wave energy: The Mutriku case study," *J. Ocean Eng. Mar. Energy*, vol. 12, no. 3, pp. 1525–1542, 2026, doi: 10.1007/s40722-026-00487-0.

[2] G. Ibarra-Berastegi, J. Sáenz, A. Ulacia, P. Serras, G. Esnaola and C. García-Soto, "Electricity production, capacity factor, and plant efficiency index at the Mutriku wave farm (2014–2016)," *Ocean Eng.*, vol. 147, pp. 20–29, 2018, doi: 10.1016/j.oceaneng.2017.10.018.

[3] Basque Government, "La planta de Mutriku, primera comercial europea que genera energía con las olas, ha producido en un año 200.000 kWh," Irekia, Aug. 2012 (in Spanish). [Online]. Available: irekia.euskadi.eus. Accessed: Oct. 26, 2026.

[4] L. Gato et al., "Sea trial results of the biradial and Wells turbines at Mutriku wave power plant," *Energy Convers. Manag.*, vol. 268, Art. no. 115936, 2022, doi: 10.1016/j.enconman.2022.115936.

[5] J. Bald et al., "The contribution of the SafeWAVE EU project to the future development of ocean energy," *Int. Mar. Energy J.*, vol. 8, pp. 73–82, 2025, doi: 10.36688/imej.8.73-82.

[6] I. Heras-Saizarbitoria, I. Zamanillo and I. Laskurain, "Social acceptance of ocean wave energy: A case study of an OWC shoreline plant," *Renew. Sustain. Energy Rev.*, vol. 27, pp. 515–524, 2013, doi: 10.1016/j.rser.2013.07.032.

[7] L. Johanning, "The fascinating history of wave energy," WEDUSEA, Jun. 3, 2025. [Online]. Available: wedusea.eu/the-fascinating-history-of-wave-energy/. Accessed: Oct. 20, 2026.

[8] IEA-OES, "First generation devices," Ocean Energy Systems, n.d. [Online]. Available: ocean-energy-systems.org/what-is-ocean-energy/waves/first-generation-devices/. Accessed: Oct. 20, 2026.

[9] International WaTERS, "Mutriku Area," European Marine Energy Centre and Ocean Energy Systems, n.d. [Online]. Available: internationalwaters.info/mutriku-area. Accessed: Oct. 26, 2026.

[10] Tethys, "Mutriku Wave Power Plant," Tethys Knowledge Base, n.d. [Online]. Available: tethys.pnnl.gov/project-sites/mutriku-wave-power-plant. Accessed: Oct. 30, 2026.

[11] L. Frayer, "Spain expands renewables with wave-powered electricity plant," NPR, Nov. 26, 2012. [Online]. Available: wfae.org/2012-11-26/spain-expands-renewables-with-wave-powered-electricity-plant. Accessed: Oct. 30, 2026.

[12] I. Muxika, P. Vinagre and J. Bald, "Deliverable 2.4: Monitoring of seafloor integrity," WESE project, Nov. 30, 2020. [Online]. Available: tethys.pnnl.gov/publications/wese-deliverable-24-monitoring-seafloor-integrity. Accessed: Oct. 30, 2026.

[13] EVE, "The Mutriku wave plant achieves cumulative electricity production of three million kilowatts per hour," Jan. 11, 2024. [Online]. Available: eve.eus/en/?p=5857. Accessed: Oct. 25, 2026.

[14] Interreg Europe, "BIMEP (Mutriku): World's first breakwater wave power plant: acceptance of the community," n.d. [Online]. Available: interregeurope.eu/good-practices. Accessed: Oct. 25, 2026.

[15] J. Lopez-Mendia, A. Aristondo, P. Ricci, J. Lekube, S. Ceballos and E. Robles, "Improving the power production of Mutriku Wave Power Plant throughout tuning the actual generator and damping valve limits," *Ocean Eng.*, vol. 315, Art. no. 119917, 2025, doi: 10.1016/j.oceaneng.2024.119917.

[16] Basque Government, "Euskadi dispone ya de la primera planta europea que suministra energía a través de las olas," Irekia, Jul. 7, 2011 (in Spanish). [Online]. Available: irekia.euskadi.eus. Accessed: Oct. 29, 2026.

[17] Offshore Energy, "Mutriku wave plant hits 2GWh mark," Offshore Energy, Feb. 24, 2020. [Online]. Available: offshore-energy.biz/2020/02/24/mutriku-wave-plant-hits-2gwh-mark. Accessed: Oct. 30, 2026.

[18] E. Le Bourhis et al., "Deliverable 2.6: Data results and analysis towards impacts' evaluation," SafeWAVE project, Dec. 17, 2024. [Online]. Available: tethys.pnnl.gov/sites/default/files/publications/Bourhis_etal_2024.pdf. Accessed: Oct. 30, 2026.